

3.1.4 Proteins

3.1.4.1 General properties of proteins

Content	Opportunities for skills development
Amino acids are the monomers from which proteins are made. The general structure of an amino acid as: $\begin{array}{c} \text{R} \\ \\ \text{H}_2\text{N} - \text{C} - \text{COOH} \\ \\ \text{H} \end{array}$ where NH_2 represents an amine group, COOH represents a carboxyl group and R represents a side chain. The twenty amino acids that are common in all organisms differ only in their side group. A condensation reaction between two amino acids forms a peptide bond. • Dipeptides are formed by the condensation of two amino acids. • Polypeptides are formed by the condensation of many amino acids. A functional protein may contain one or more polypeptides. The role of hydrogen bonds, ionic bonds and disulfide bridges in the structure of proteins. Proteins have a variety of functions within all living organisms. The relationship between primary, secondary, tertiary and quaternary structure, and protein function. The biuret test for proteins. Students should be able to relate the structure of proteins to properties of proteins named throughout the specification.	AT f Students could use, and interpret the results of, a biuret test for proteins. AT g Students could use chromatography with known standard solutions, to separate a mixture of amino acids and identify their components.

3.1.4.2 Many proteins are enzymes

Content	Opportunities for skills development
Each enzyme lowers the activation energy of the reaction it catalyses. The induced-fit model of enzyme action. The properties of an enzyme relate to the tertiary structure of its active site and its ability to combine with complementary substrate(s) to form an enzyme-substrate complex. • The specificity of enzymes • The effects of the following factors on the rate of enzyme-controlled reactions – enzyme concentration, substrate concentration, concentration of competitive and of non-competitive inhibitors, pH and temperature. Students should be able to: • appreciate how models of enzyme action have changed over time • appreciate that enzymes catalyse a wide range of intracellular and extracellular reactions that determine structures and functions from cellular to whole-organism level.	MS 0.5 Students could be given the hydrogen ion concentration of a solution in order to calculate its pH, using the formula: $pH = -\log_{10}[H^+]$

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Required practical 1: Investigation into the effect of a named variable on the rate of an enzyme-controlled reaction.	PS 2.4 Students could identify the variables that must be controlled in their investigation into rate of reaction. PS 3.3 Students could calculate the uncertainty of their measurements of the rate of reaction. MS 3.2 Students could select an appropriate format for the graphical presentation of the results of their investigation into the rate of enzyme-controlled reactions. MS 3.6 Students could use a tangent to find the initial rate of an enzyme-controlled reaction.